

34. LSTM-Based E-commerce Sales Prediction

Book Chapter Title: *Enhancing E-commerce Sales Prediction: Overcoming Data and Complexity Challenges with LSTM Models (Accepted for Publication)*

CRC Press, Taylor & Francis Group, USA: *Intelligent Systems: Emerging Trends and Challenges in Deep Neuro Fuzzy Models and Explainable Artificial Intelligence*

Authors: Sangita Dutta, Dr. Siddhartha Roy

35. Interdisciplinary Advances in AI

Book Nova Science Publisher: *Intelligent Systems and Emerging Technologies: Interdisciplinary Advances in AI, Engineering, and Computational Modeling (Accepted for Publication)*

Authors: Dr. Siddhartha Roy, Dr. Malay Gangopadhyaya, Dr. Sandip Mandal

36. Tele-Oncology in Bioinformatics

Chapter Title: *An Intervention of Tele-Oncology in Palliative Care: A Critical Review and Analysis of Remote Monitoring, Cancer Diagnosis, and Treatment (Accepted for Publication)*

Apple Academic Press (CRC Press): *Intelligent Applications of Bioinformatics: A Practical Approach*

Authors: Partha Sarker, Dr. Siddhartha Roy

37. AI in Drug Discovery and Telemedicine

Chapter Title: *Machine Learning Transformations in Drug Discovery: A Paradigm Shift in Development Strategies (Accepted for Publication)*

De Gruyter Publishers (Scopus Indexed): *Drug Discovery and Telemedicine Through Artificial Intelligence, Computer Vision, and IoT*

Author: Dr. Siddhartha Roy

Awards & Recognitions

1. 🏆 Aegis Graham Bell Award Finalist (Feb 2025), Supported by Govt. of India

Project: AquaSmartIQ – A smart Aquaculture monitoring platform using IoT and AI

🔍 Team: Apurba Nandi, Avik Kumar Das, Sandip Mandal, Arijeet Ghosh



Image 1: University of Engineering & Management (UEM), was recognized as a finalist under the 'Innovation in RPA' category and **Dr. Sandip Mandal**, Head of the Department, CSE (IoT, CS & BT), received the award in the esteemed presence of **Hon'ble Minister Shri Jitin Prasada**, Union Minister of State for Commerce and Industry, and Electronics and Information Technology, Government of India.



Image 2: The award was graciously conferred upon our esteemed Vice Chancellor Sir, **Dr. Sajal Das Gupta**, accompanied by team members **Avik Kumar Das** and **Apurba Nandi** in recognition of their contribution.

2. 🏆 2nd Runner Up – CMPDIL Hackathon 2025, Coal India Ltd. Govt. of India.

Project: Intelligent Fire Monitoring & Ventilation Control System for Underground Mines

👥 **Team:** Dr. Sandip Mandal, Avik Kumar Das, Apurba Nandi, Koustov Mondol, Dr. Sajal Dasgupta

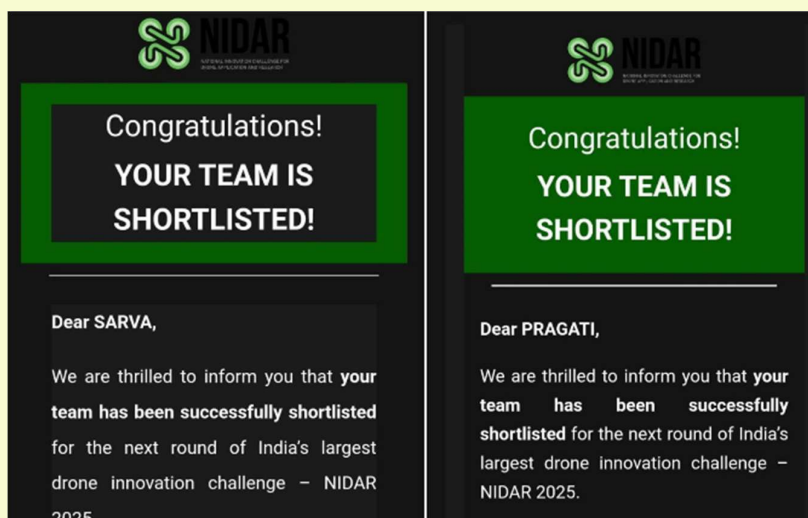
Image 1: Our team secured the 2nd Runner-Up position for *Problem Statement 3: Intelligent Monitoring and Control System for Ventilation and Fire Management in Underground Coal Mines* at Hackathon on R&D 2025, organized by CMPDI (CIL) with Ministry of Coal, winning a prize of ₹2,00,000 for innovation excellence.

Problem Statement-3: Indigenous development of Intelligent Monitoring and Control System for Ventilation and Fire Management System in Underground Coal Mines		Main Prizes
IIT-Ropar, IIT-Kharagpur, IIT-Delhi and Ingenisys Integration LLP, Kolkata.	Winner	 1st Prize INR 4,00,000
NIT, Rourkela	1st Runner Up	 2nd Prize INR 3,00,000
University of Engineering and Management, Kolkata.	2nd Runner Up	 3rd Prize INR 2,00,000

3. Two innovation-driven student teams — **Team Sarva** and **Team Pragati** — have selected to the next round of **NIDAR 2025**, India's **premier National Innovation Challenge** for Drone Applications.

Team Sarva, led by **Yashraj Panda**, is addressing critical challenges in Disaster Management, while Team Pragati, guided by **Tuheena Bose**, is transforming Precision Agriculture through smart drone technology.

With the unwavering support of faculty mentors **Prof. Apurba Nandi** and **Prof. Avik Kumar Das**, these teams exemplify the spirit of impactful, real-world innovation and continue to push boundaries in drone research and application.



4. 🏆 **Team 18 won the JPMorgan Chase Code for Good 2025 Hackathon** held in Mumbai on June 7–8! Among the standout contributors was **Hritajit Sur** (3rd from left in Image 1), a brilliant student from our **IEDC Lab, UEM Kolkata**, who helped build **FootPathshala** — a smart, field-ready solution for UPAY that automates attendance, assessments, curriculum, and volunteer tracking. Kudos to JPMorgan Chase for this inspiring platform and to UEM for nurturing future-ready talent!



Industry-Academic Collaboration

1. BIRAC-Funded Project

Collaborating institutions: Regional Institute of Ophthalmology (RIO), Indian Institute of Science (IISc), UEM

🔍 **Leads:** Apurba Nandi, Avik Kumar Das

📄 **Output:** Copyrighted OCT dataset, AI models for diabetic retinopathy diagnosis

2. Research Collaboration with Jadavpur University

📄 **Outcome:** Publication in *Journal of Cognitive Neurodynamics* (🧠 Impact Factor: 3.10)

3. Ongoing Project with IEST, Shibpur

🔍 **Focus Area:** Research on underwater technologies

📍 **Outcome:** Active interdisciplinary research in progress

4. Collaboration with WEBEL

🏆 **Outcome:** Industrial training provided to lab students, strengthening industry-readiness and practical exposure

Product Development Highlights

- 📄 **2 Copyrights Filed:**
 - OCT Image Dataset – For retinal disease detection (BIRAC collaboration)
 - Temple Architecture Dataset – For cultural heritage preservation and classification
- 🔒 **12 Patents Filed in Diverse Domains:**
 - Agriculture – Smart irrigation, disease detection
 - Medical Diagnostics – AI-based early disease screening